Wine Logistical Regression Report

1.0 Exploratory Data Analysis

1.1 Missing Values

The dataset arrived with 8,200 missing values spread across nine variables. No single variable was missing enough data to justify removal, the worst offender, `STARS`, was missing approximately 26% of its records. For the sole categorical variable requiring imputation (`STARS`), missing values were replaced with the mode (2 stars), as this is the most frequently occurring rating and the most defensible substitute for an ordinal scale. The seven continuous variables with missing data were imputed using each variable's median rather than the mean, as several variables exhibited skewness and extreme values that would distort a mean-based imputation.

1.2 Response Variable Balance

Figure A1 shows the distribution of the `Purchase` outcome. The response variable is notably imbalanced: 78.6% of records (10,061) represent a purchase, with only 21.4% (2,734) representing non-purchases. This mirrors commercial reality, with most wines presented to buyers are eventually sold, but it carries important modeling implications. A naive classifier that simply predicts "purchase" for every record would still achieve 78.6% accuracy, making raw accuracy a misleading performance metric. Model evaluation should therefore prioritize measures such as AUC, sensitivity, and specificity over accuracy alone.

1.3 Outliers and Data Quality

Outlier detection using the standard IQR method (1.5 × IQR beyond the first and third quartiles) revealed a pervasive and serious data quality problem. Nearly every continuous variable contained physically impossible values, negative alcohol content, negative residual sugar, negative sulfur dioxide measurements, and pH values below 1.0. These are not legitimate measurements; they reflect upstream data entry or instrument errors.

Given that outlier rates reached 33% in some variables, removing affected rows entirely was not viable, a row-deletion approach applied across all 11 continuous variables simultaneously would have eliminated most of the dataset through compounding exclusions. Instead, winsorization was applied: extreme values were capped at the IQR fence boundaries ($Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$) rather than deleted. This approach retained all 12,795 records while preventing implausible values from distorting model coefficients and is the standard practice when outlier contamination is widespread rather than isolated.

1.4 Correlations Among Variables

A full correlation heatmap reveals two noteworthy inter-predictor relationships. `Cases` and `LabelAppeal` share a moderate positive correlation ($r = 0.36$), and `STARS` and `LabelAppeal` are also modestly correlated ($r = 0.29$), suggesting that wines with more attractive labels tend to

receive better expert ratings. `FixedAcidity` and `AcidIndex` show a positive relationship (r = 0.19), which is expected since `AcidIndex` is a composite acidity measure. Crucially, the chemistry variables show almost no meaningful correlation with one another beyond these exceptions, with the vast majority of pairwise values falling between -0.05 and 0.05. This is a favorable finding for logistic regression, as it suggests multicollinearity will not be a significant problem among the predictors entering the model.

1.5 Correlations with Purchase

Figure A2 ranks each variable's linear association with `Purchase`. The most notable finding is the strong positive correlation between `Purchase` and `Cases` (r = 0.82), which appears visually dominant in the matrix. This reflects data leakage as `Cases` records how many cases were ordered, making it a consequence of the purchase decision rather than a predictor of it. Thus, excluding it from all models.

After excluding `Cases`, the two most meaningful predictors are `AcidIndex` (r = -0.27) and `STARS` (r = 0.17). The negative relationship with `AcidIndex` indicates that wines perceived as more acidic are less likely to be purchased, a finding consistent with consumer taste preferences. The positive relationship with `STARS` confirms that professional quality ratings meaningfully influence buying decisions. All remaining variables had absolute correlations below 0.10, suggesting that wine chemistry alone carries limited linear predictive power. This does not preclude their usefulness in a model, as logistic regression can extract signal from modest predictors in combination, but it does set realistic expectations about model performance.

2.0 Logistical Regression Model

2.1 Pre-Model Preparation

Before any model could be estimated, one additional data preparation step was necessary beyond what was performed during EDA. The original `AcidIndex` variable spanned 14 distinct levels (4 through 17), and the frequency table from the EDA revealed severe sparsity at the high end of the scale. To address this, the 14 original levels collapsed into five meaningful groups prior to dummy variable creation, shown in Table 1 below.

Table 1. `AcidIndex` *Groupings*

Group	Original Levels	Rationale
Low	4, 5, 6	Below-average acidity, sparse at floor
Mid_Low	7	First well-populated level
Mid	8	Modal category
Mid_High	9	Above-average acidity
High	10-17	All sparse high-acidity levels collapsed

The grouping followed the natural distribution of the data, with breakpoints placed where population density shifted meaningfully. The Mid group (`AcidIndex` = 8) is the modal category and serves as a natural center point, while the High group consolidates all levels where

individual cell counts were too small to support reliable estimation. This reduced the `AcidIndex` contribution from 13 dummy variables to 4, eliminating sparsity while preserving the ordinal structure of the variable.

2.2 Model Selection Strategy

Three automated selection procedures were evaluated to identify the best predictive model – backwards elimination, forward selection, and stepwise selection. All three methods used AIC as the selection criterion, which balances model fit against complexity by penalizing the addition of unnecessary predictors. The full model containing all available predictors served as the upper bound, and a null intercept-only model served as the lower bound.

The most noteworthy finding from this process is that all three selection procedures converged on the exact same 16-predictor model with an identical AIC of 7,524.5 and residual deviance of 7,490.5. This agreement across methods that approach variable selection from opposite directions provides strong evidence that this variable set represents a genuine optimum rather than an artifact of any single selection path.

Compared to the full model (AIC 7,530.4), the selected model eliminated six variables, `ResidualSugar`, `Chlorides`, `Density`, `Alcohol`, `LabelAppeal_-1`, and `AcidIndex_grouped_Mid_Low`, each of which contributed insufficient explanatory power to justify inclusion. Their removal improved AIC by 5.9 points while costing only 6 degrees of freedom, indicating a favorable trade. Table 2 below summarizes the model comparison resulting in the final model.

Table 2. Model Comparison Summary

Model	Predictors	Residual Deviance	AIC
Null	0	9338.3	9340.3
Full	22	7484.4	7530.4
Final Model	16	7490.5	7524.5

2.3 Coefficient Interpretation

The final model selected by all three automated procedures estimates the log-odds of wine purchase as a linear combination of 16 predictors spanning wine chemistry, expert ratings, label aesthetics, and composite acidity grouping. Formally written as:

logit(P(`Purchase` = 1)) = 2.541 − 0.009(`FixedAcidity`) − 0.242(`VolatileAcidity`) + 0.117(`CitricAcid`) + 0.001(`FreeSulfurDioxide`) + 0.001(`TotalSulfurDioxide`) − 0.163(`pH`) − 0.186(`Sulphates`) − 0.530(`STARS_2`) + 17.18(`STARS_3`) + 17.29(`STARS_4`) − 0.153(`LabelAppeal_0`) − 0.426(`LabelAppeal_1`) − 0.727(`LabelAppeal_2`) − 0.224(`AcidIndex_Mid`) − 0.910(`AcidIndex_Mid_High`) − 1.778(`AcidIndex_High`)

The intercept of 2.541 represents the log-odds of purchase for the baseline wine (a wine with average chemistry, a 1-star rating, the least appealing label (−2), and low acidity), which

translates to a baseline purchase probability of approximately 92.7%. This high baseline reflects the overall class imbalance in the data where 78.6% of wines were purchased.

The acidity variables collectively form the strongest and most internally consistent signal in the model. `AcidIndex_grouped_High` is the single most influential predictor (β = -1.778, OR = 0.169), indicating that wines in the highest acidity group have approximately 83% lower odds of being purchased compared to the Low acidity reference, holding all other variables constant. The progression across acidity groups is clean and monotonic, exactly what one would expect from consumer behavior, as wines become increasingly acidic, buyers become progressively less likely to purchase them. `VolatileAcidity` (OR = 0.785) reinforces this narrative as high volatile acidity produces vinegar-like off-flavors that are a recognized quality defect in wine, and buyers appear to respond to this accordingly. `pH` (OR = 0.850) contributes a further acidity signal, though its interpretation requires care as extreme pH values in either direction typically indicate quality abnormalities rather than a straightforward acidity preference.

`STARS_2` (OR = 0.589) is statistically significant, indicating that 2-star wines have 41% lower odds of purchase than 1-star wines holding other variables constant. This counterintuitive direction likely reflects a data artifact where the 1-star reference group is small and may represent niche wines with committed buyers who are undeterred by low expert ratings. `STARS_3` and `STARS_4` both produced coefficients of approximately 17 with standard errors exceeding 150 and p-values near 1.0, signifying a quasi-complete separation. Within the training data, nearly every 3-star and 4-star wine was purchased, leaving insufficient non-purchase cases for stable estimation. These predictors are retained in the model because their presence improves overall fit, but their individual coefficients carry no interpretive value. Collapsing `STARS_3` and `STARS_4` into a single high-rating category is the recommended remedy in future iterations.

The `LabelAppeal` coefficients all carry negative signs relative to the -2-reference level, with increasing magnitude as labels become more visually appealing: `LabelAppeal_0` (OR = 0.858), `LabelAppeal_1` (OR = 0.653), and `LabelAppeal_2` (OR = 0.483). This suggests that more attractive labels are associated with lower purchase probability, the opposite of conventional marketing wisdom. The explanation may lie in the unusual reference category. Wines rated -2 on label appeal represent only 3.9% of the dataset, indicating an extremely small, specialized group that buyers apparently sought out regardless of unappealing packaging, possibly boutique or cult wines where aesthetics are irrelevant. With such an atypical baseline, the apparent negative direction of these coefficients is a function of the reference group rather than a genuine statement that attractive labels reduce purchase likelihood. `LabelAppeal` contributes statistically significant predictive information but should be interpreted contextually rather than literally.

2.4 Goodness-of-fit

The likelihood ratio test is the most fundamental goodness of fit assessment for logistic regression. It compares the fitted model to the null model, one that knows nothing about the predictors and simply predicts the overall purchase rate for every observation. The test statistic of

$\chi^2(16) = 1{,}847.8$ with $p < 0.0001$ provides overwhelming evidence that the 16 predictors collectively explain significantly more variation in purchase behavior than chance alone. The model reduced deviance from 9,338.3 (null) to 7,490.5 (final), a 19.8% reduction. Crucially, this reduction was achieved with only 16 degrees of freedom consumed out of 8,989 available, meaning the model achieved substantial fit improvement at very low cost in complexity. The residual deviance of 7,490.5 on 8,973 degrees of freedom yields a ratio close to 1.0 (7,490.5 / 8,973 = 0.835), which is favorable.

2.5 Test Set Performance & Diagnostics

Several validation performance metrics were calculated on the test set of data that was left out during the modeling development to understand the accuracy of the final model. Table 3 reports the confusion matrix from the test set. This represents the distribution of the true negatives (TN), false positives (FP), false negatives (FN), and true positives (TP). Table 4 indicates the sensitivity/recall, specificity, accuracy, precision, and F1 score – all derived from the confusion matrix. As a final performance check, the Area Under the Curve (AUC) was calculated and a Receiver Operating Characteristic (ROC) curve was generated to illustrate the performance of the model using the sensitivity rate over specificity rate seen in figure A3.

Table 3. Confusion Matrix of Test Set

	Predicted: Not Purchased	**Predicted: Purchased**
Actual: Not Purchased	188 (TN)	633 (FP)
Actual: Purchased	91 (FN)	2,893 (TP)

Table 4. Test Set Performance Metrics

Metric	**Training Set**	**Test Set**
Observations	8,990	3,805
Accuracy	-	81%
Sensitivity (TP rate)	-	96.8%
Specificity (TN rate)	-	22.9%
Precision	-	82.3%
F1 Score	-	0.889
AUC	-	0.778

The AUC of 0.778 confirms meaningful discriminatory ability, the model correctly ranks a randomly selected purchaser above a randomly selected non-purchaser 77.8% of the time, substantially above the 0.50 random baseline. The close alignment between training deviance and test set performance suggests the model is not overfit to the training data, which is expected with the large sample size relative to the number of predictors. However, the performance profile reveals a fundamental tension created by the class imbalance. The sensitivity of 96.8% reflects strong identification of actual purchases, while the specificity of 22.9% means the model misclassifies approximately 77% of actual non-purchases as predicted purchases. The confusion

matrix makes this concrete: of 821 actual non-purchases in the test set, the model correctly identified only 188 while misclassifying 633 as purchases.

This asymmetry is a direct consequence of the 78.6% purchase rate in the data; the model learns that predicting purchase is almost always correct and becomes overly conservative about predicting non-purchase. At the default 0.50 threshold, the model is optimized for sensitivity at the expense of specificity. Raising the threshold to 0.65 or 0.70 would rebalance this tradeoff, improving the identification of non-purchases at a modest cost to purchase identification. The right threshold depends on the business cost structure: if the cost of stocking unsold inventory exceeds the cost of declining a potential sale, a higher threshold is warranted, and that determination belongs with business stakeholders rather than in the model itself.

3.0 Conclusions

The most consequential finding from this analysis is how few variables drive wine purchase decisions in a meaningful way. Of the fourteen candidate predictors entering the model, the story is reduced to two themes: acidity and credibility. Wines that score higher on composite acidity measures are consistently and dramatically less likely to be purchased, with high-acidity wines facing 83% lower odds of purchase compared to their low-acidity counterparts. Expert ratings, captured through `STARS`, represent the primary positive signal, where professional quality assessments matter more to buyers in a measurable way. Everything else (i.e., residual sugar, alcohol content, density, and chlorides) contributed little predictive power that the selection algorithms discarded them entirely. Sulfur dioxide variables surviving the model with positive coefficients is also worth noting. Rather than reflecting a direct taste preference, their significance most plausibly signals that wines with well-managed preservation chemistry are better-produced wines overall. It is a proxy for winemaking competence rather than a sensory driver.

Before any modeling could begin, a substantial data cleaning effort was required. Negative values appeared across variables where negative measurements are physically impossible, such as negative alcohol content, negative residual sugar, pH values below 1.0. Outlier rates reached 33% in some variables. The sheer scale of data quality problems suggests that measurement and recording standards at the point of data collection are inconsistent at best. For an industry that prides itself on precision and craft, this is a notable gap between the production floor and the analytical record. Any organization relying on this data for decision-making without first auditing its quality would be building strategy on a flawed foundation.

For winemakers, the acidity findings carry a direct actionable message: perceived acidity is the single most controllable driver of commercial outcomes in this dataset. Winemakers who optimize for critical scores or competition awards while neglecting accessible, lower-acid profiles may be producing technically excellent wines that nonetheless face commercial headwinds. For distributors and buyers, the strong relationship between STARS and purchase behavior suggests that expert rating systems carry genuine commercial weight beyond marketing

as they appear to influence actual purchasing decisions in the data. Prioritizing relationships with well-rated producers and using ratings as a procurement filter is supported by the evidence here.

Several aspects of this analysis challenged assumptions coming in. The discovery that all three selection methods — backwards, forward, and stepwise — converged on identical variables was the most instructive moment in the modeling process. That convergence also reinforced a broader lesson: when the underlying signal in data is real and clean, analytical methods tend to find it regardless of the path taken.

The class imbalance problem and its consequences for model evaluation was a second important lesson. An 81% accuracy figure sounds impressive until one recognizes that predicting "purchase" for every single record would achieve 78.6% accuracy with no model at all. Sensitivity of 96.8% paired with specificity of 22.9% tells a far more honest story that the model is excellent at confirming what is likely to sell and poor at identifying what is not. This asymmetry is not a failure of the modeling procedure, but rather a faithful reflection of an imbalanced world. It also means that deploying this model to support inventory decisions without threshold adjustment would create systematic blind spots around non-purchase predictions.

A. Appendix

Figure 1. Response Variable (Purchase) Distribution

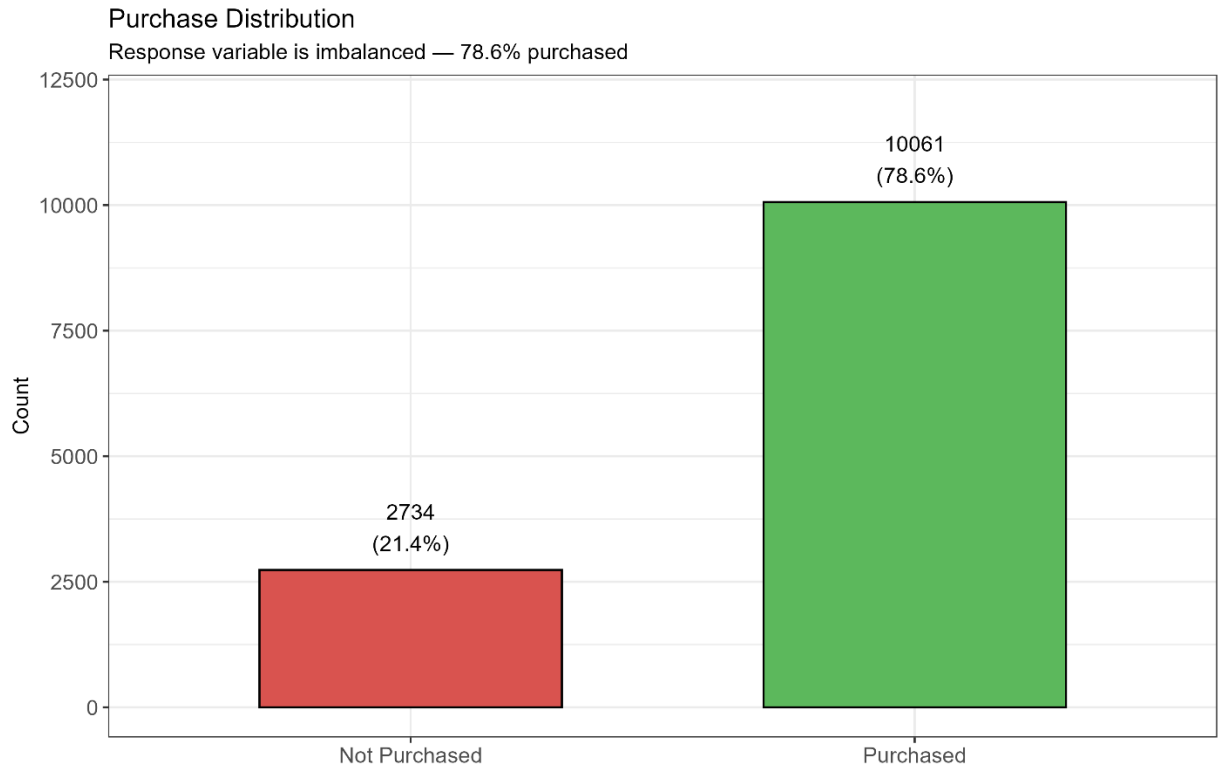

Figure 2. Correlation Bar Chart with Response Variable

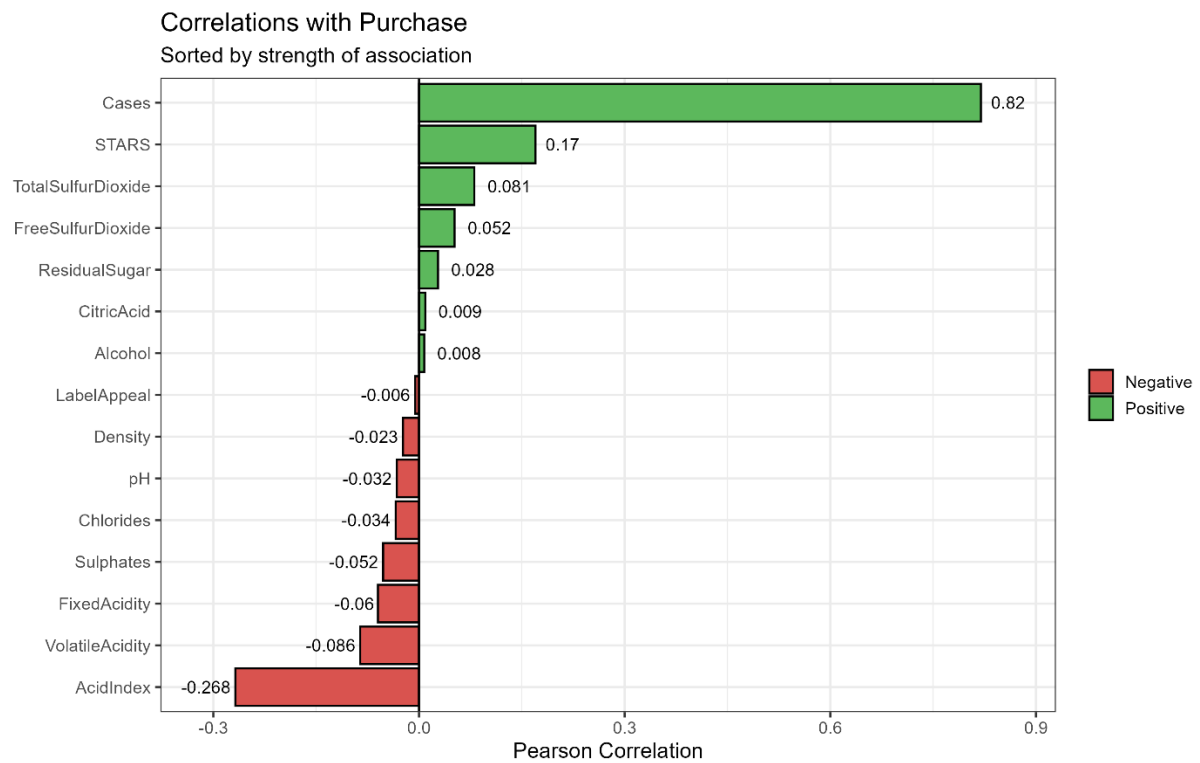
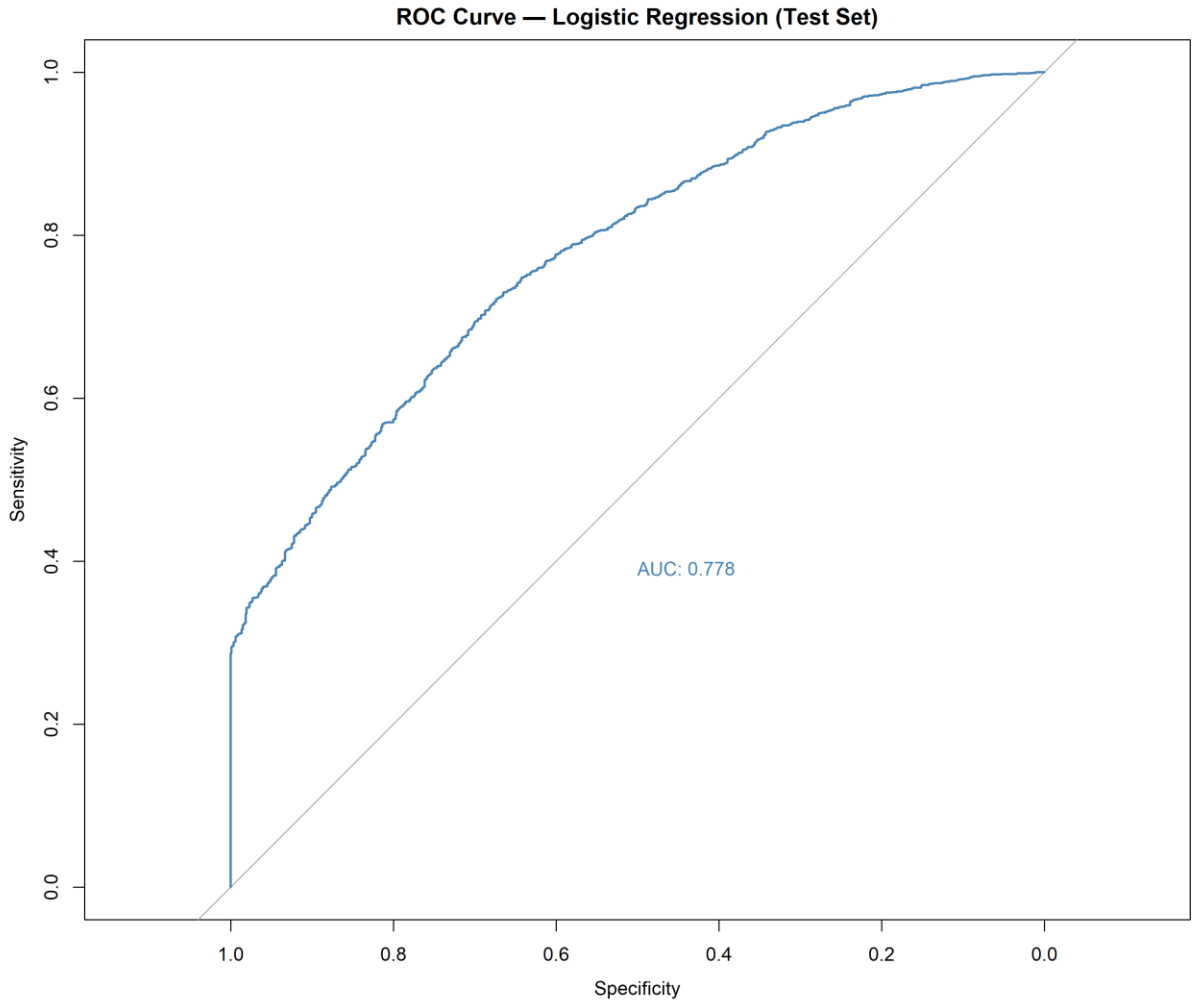
Figure 3. ROC Curve for Logistic Regression on Test Set